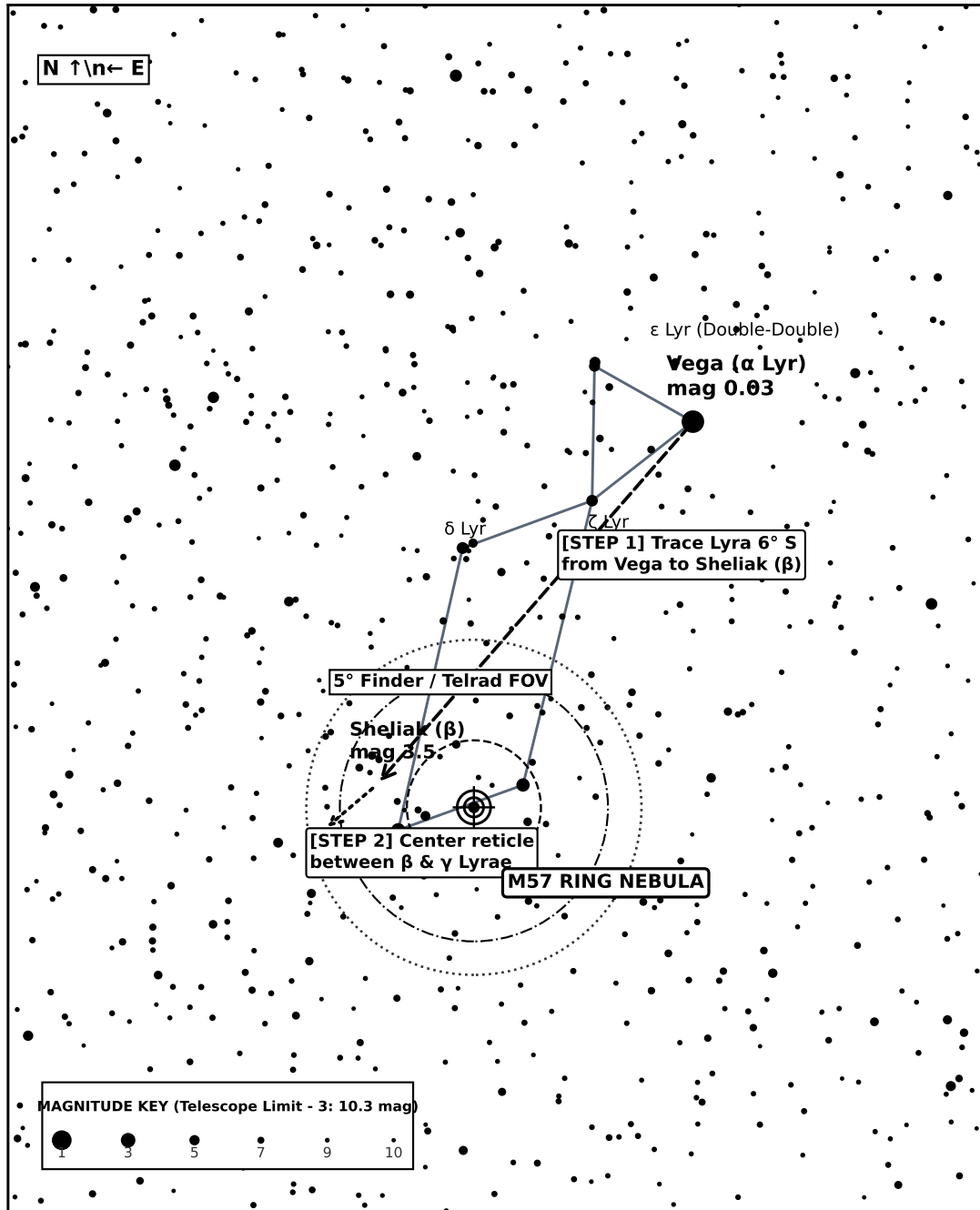


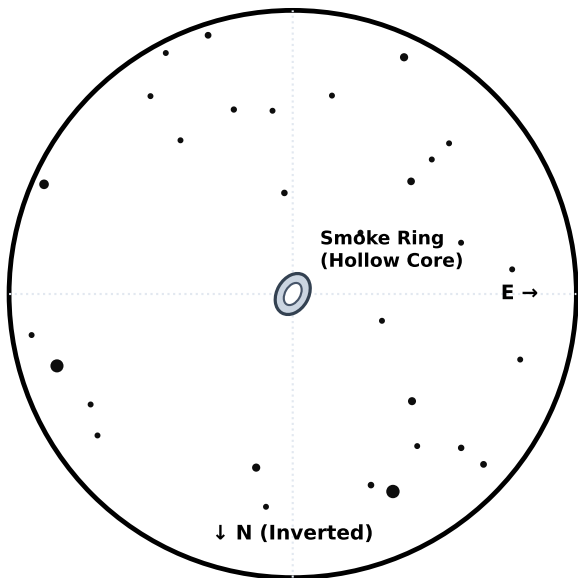
STAR-HOPPING FINDER CHART 1: M57 (RING NEBULA) — LYRA

Location: Dvigrad (Bortle 4, SQM 21.0) | Sep 12, 2026 • 21:15 CEST | Sky-Watcher 200P Dobsonian (8-inch f/6, 1200mm FL) | B/W Toner-Saver Edition

VIEWPORT A: WIDE-FIELD STAR-HOPPING CHART (Upright Naked-Eye / Finder: N ↑, E ←)



VIEWPORT B: EYEPIECE SIMULATION (Negative Inverted 180°: N ↓, E →)



12.5mm Eyepiece (96x Magnification, 31' True FOV)

VIEWPORT C: TARGET DOSSIER & STAR-HOPPING INSTRUCTIONS

OBJECT DOSSIER: M57 (NGC 6720)

- Constellation: Lyra | Type: Planetary Nebula
- Magnitude: 8.8V | Surface Brightness: 9.3 mag/arcmin²
- Apparent Size: 1.4' × 1.0' | Distance: ~2,570 light-years
- Tonight's Ephemeris at Dvigrad (Sep 12, 2026):
 - 20:00 CEST: Alt 76.4°, Az 204° (South-Southwest)
 - 21:15 CEST: Alt 75.4°, Az 218° (Zenith Sweet Spot!)
 - 22:30 CEST: Alt 64.3°, Az 237° (Safe > 60° Alt)

OBSERVING STRATEGY & EYEPIECES:

- Finder (20mm / 60x): M57 shows as a tiny, unmistakable ghostly out-of-focus disk among sharp field stars.
- Target (12.5mm / 96x): Resolves the smoke ring shape! Averted vision clearly reveals the hollow dark core.
- Filter: Under Dvigrad's Bortle 4 skies, radiant without filters.

STEP-BY-STEP STAR HOP NARRATIVE:

1. Naked-Eye: Locate brilliant Vega (mag 0.0) near Zenith.
2. Identify Lyra's parallelogram ~6° south of Vega. Find the two base stars: Sheliak (β, mag 3.5) and Sulafat (γ, mag 3.2).
3. Slew Telrad / Finder: Place central reticle between β and γ, slightly closer to Sulafat (about 60% of the distance).
4. Eyepiece: Look for the ghostly smoke-ring between two stars!